



Project HAMi

让异构算力因开源而好用
最大化利用率，为效率而生

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DYNAMIA

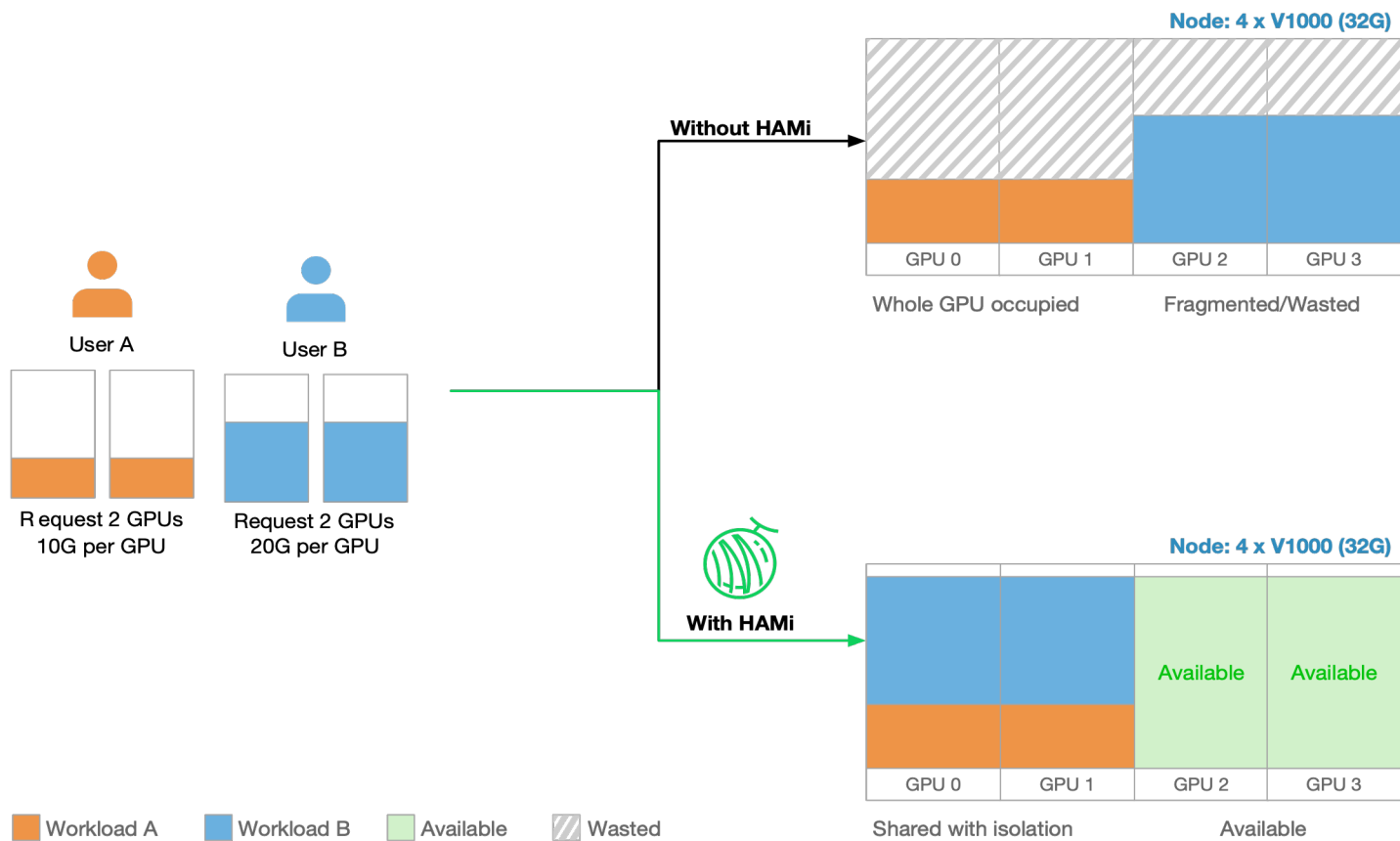
目录



-  HAMi介绍
-  HAMi v2.9新功能
-  DRA生态联盟

HAMi > Kernel

Kernel为做GPU时代的操作系统



GPUS 需求4 → 2

利用率45% → 90%



异构算力必然成为主流

国内算力的发展是基于国家战略和商业决策共同决定的

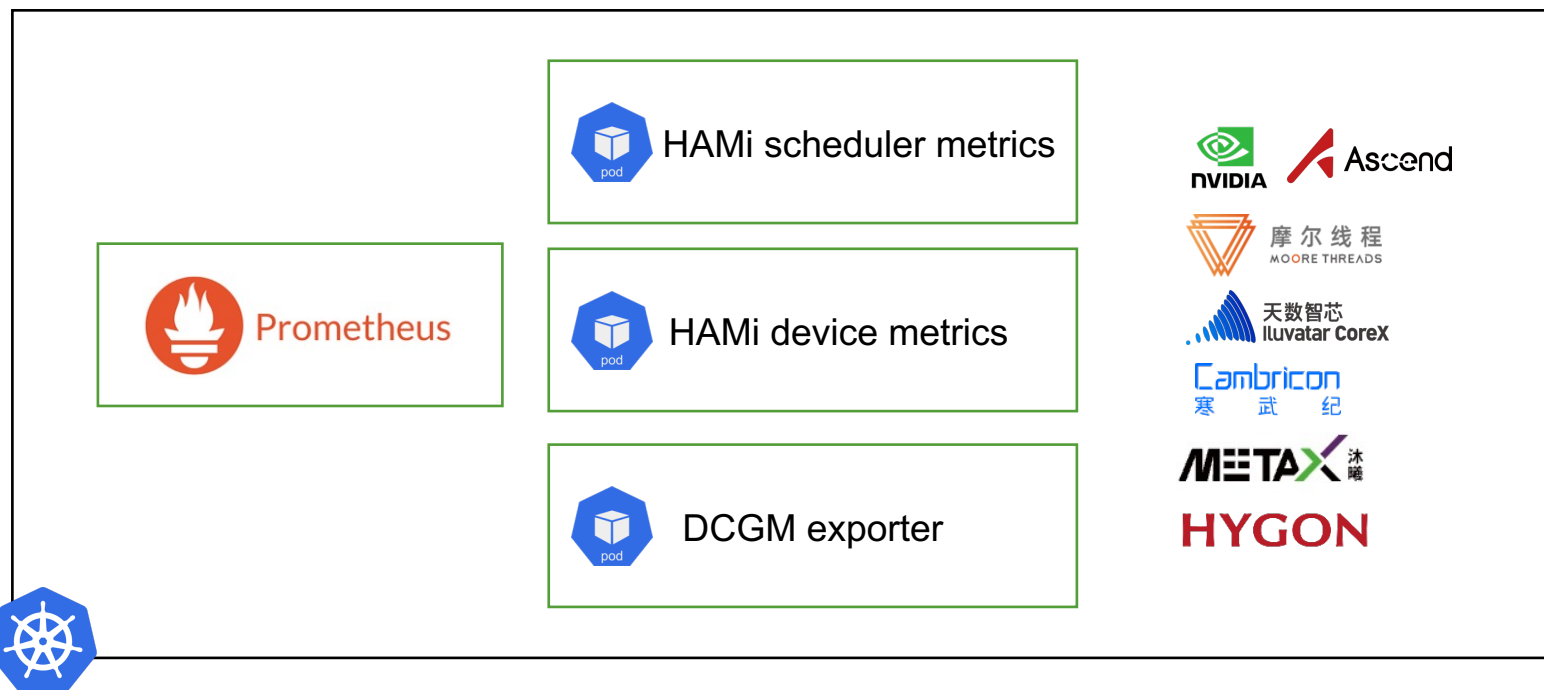
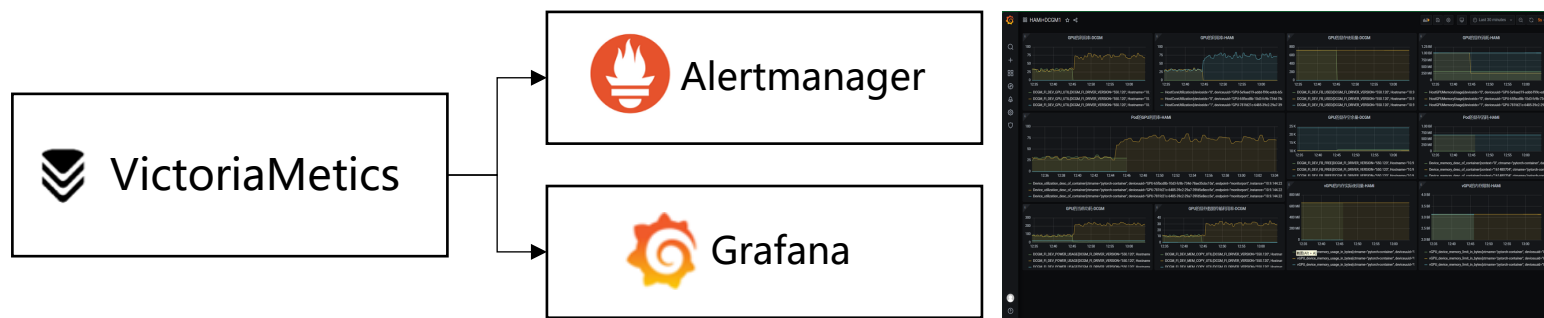


商业 + 战略，双驱动



HAMi可观测性 > 调度和分配

真实使用尽在掌握



01

K8s 调度维度: vGPU 都被哪些任务使用了, 任务、GPU 绑定关系

02

节点有三张卡, 每张卡被哪些任务占用、占用了多少算力、显存

03

GPU 设备维度: 反映应用运行过程中真实算力、显存使用率

04

A100 有三个任务, 每个任务真实使用了 20% 算力(时间片)、30% 显存

05

HAMi 也提供社区最佳实践 监控面板, 一站式观测调度与真实使用情况

各行各业的大模型应用 (400+)

The diagram illustrates the AI ecosystem landscape, categorized into three main sections: AI Applications, AI Models, and AI Infrastructure.

AI Applications (Left Column):

- AI Agents:** 豆包 (Dou Bao), Dify.AI, ...
- AI Chatbots:** 百度文心一言 (Baidu Wenxin Yiyang), 智谱·AI (Zhi Pu AI), 百川智能 (BAICHUAN AI), 阿里云 Qwen (Alibaba Cloud Qwen)

AI Models (Center Column):

- Large Language Models:** 昇思 MindSpore (Sheng Si), 华为盘古大模型 (Huawei Pangu Large Model)
- Small Language Models:** 晟思 MindAIe (Sheng Si)

AI Infrastructure (Right Column):

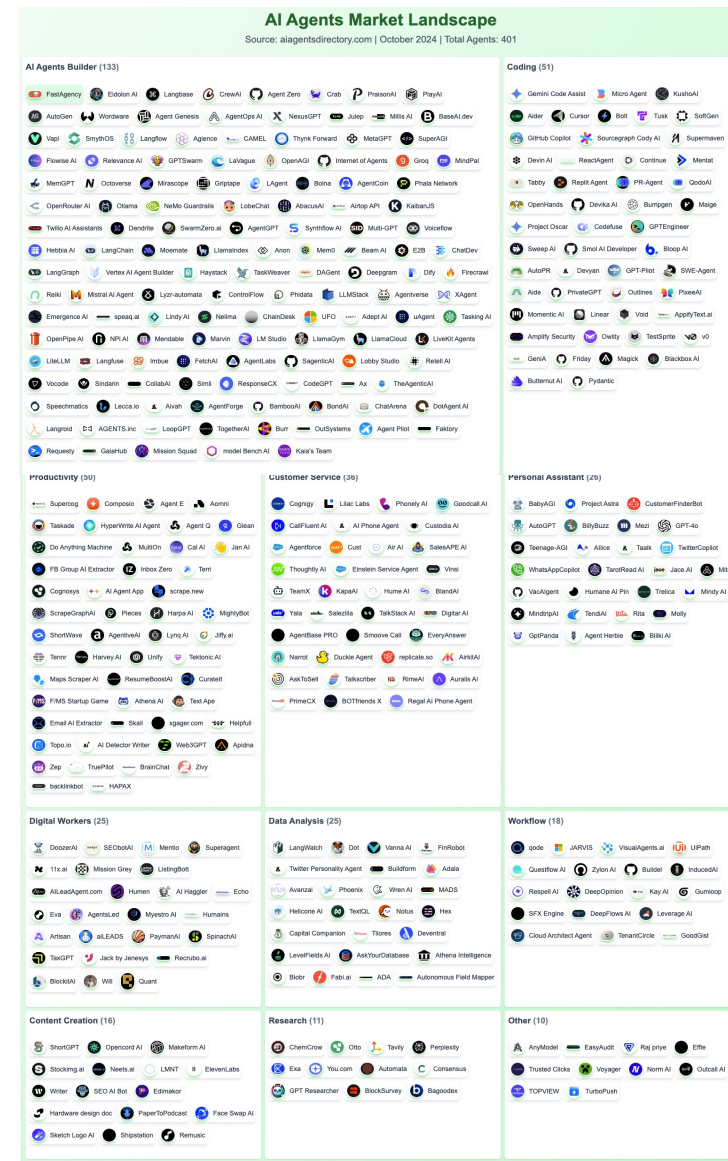
- Industry Standard APIs:** Industry Standard APIs
- Nvidia Models:** Nvidia Models
- Triton Inference Server:** Triton Inference Server
- TensorRT LLM:** TensorRT LLM
- Pytorch:** Pytorch

The diagram illustrates the integration of various AI models into the LLM ecosystem. It is divided into two main sections by a vertical dashed line. The left section, labeled 'LLM Ecosystem', contains five boxes representing different AI models: Baidu Wenxin (百度文心一言), Zhipu AI ChatGLM (智谱·AI ChatGLM), Baichuan AI (百川智能 BAICHUAN AI), Alibaba Qwen (阿里云 Qwen), and Huawei Pangu (华为云 盘古大模型). The right section, labeled 'Nvidia Models', contains two boxes representing Nvidia's models.

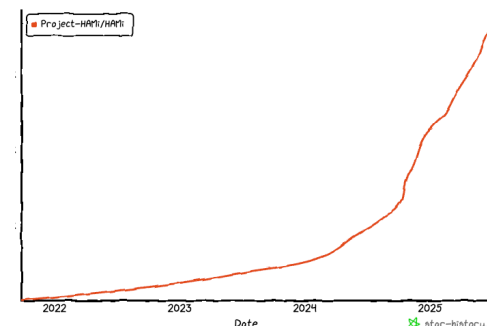
The diagram illustrates the inference stack components, separated by a dashed line. On the left side (client-side), the components are: **INFINIGENCE** (无问芯穹), **Colossal AI**, and **Pytorch**. On the right side (server-side), the components are: **晟思 MindIE**, **Triton Inference Server**, **TensorRT LLM**, and **Pytorch**.




Accelerator	Compatibility
天数智芯 Iluvatar CoreX	兼容 CANN
METAX 沐曦	兼容 CANN
摩尔线程 Moore Threads	兼容 CANN
寒武纪 Cambricon	兼容 CANN
昇腾 Ascend	兼容 CANN
CUDA	兼容 CUDA
Nvidia GPU	兼容 CUDA



全球唯一&生态初步形成

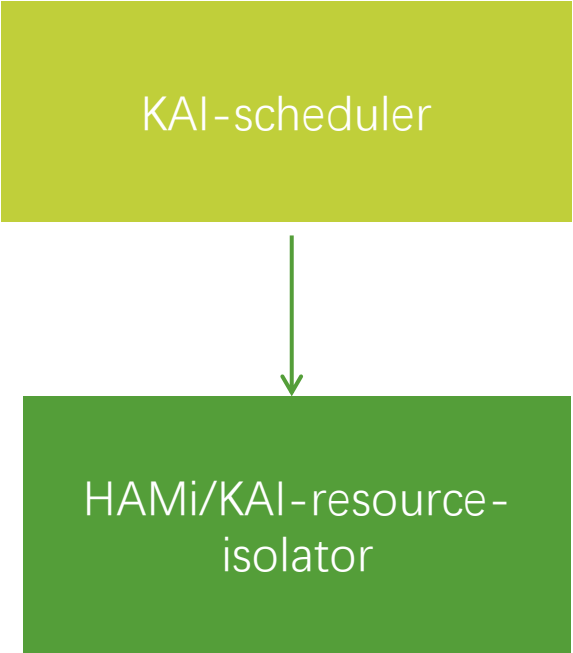


A person is walking away from the camera on a dirt path that winds through a dense bamboo forest. The bamboo stalks are tall and thin, creating a vertical rhythm in the scene. The lighting is soft and diffused, suggesting an overcast day or a shaded forest interior. The person is wearing a light-colored shirt, dark shorts, and a backpack.

HAMi的目标

HAMi 新版 本功能预热

Design: <https://github.com/kai-scheduler/KAI-Scheduler/pull/60>



KAI-scheduler

HAMi/KAI-resource-isolator

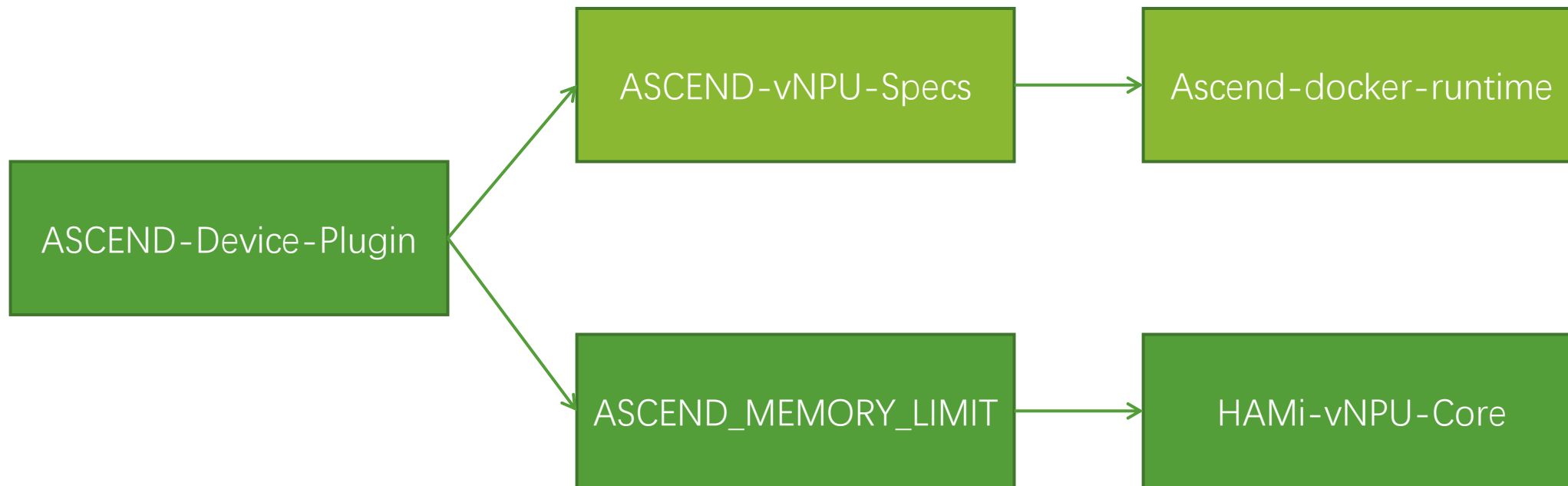
Volcano-vgpu-device-plugin 全面升级

支持pass-device-spec参数配置

适配nvidia-device-plugin 0.19.0新API

支持CDI模式

昇腾用户态切分方案（支持910C）



A person is walking away from the camera on a narrow path that winds through a dense bamboo forest. The bamboo stalks are tall and thin, creating a natural archway over the path. The lighting is soft and diffused, suggesting a misty or overcast day. The person is wearing a light-colored shirt and dark shorts, and is carrying a backpack.

HAMi的愿景

A solid green vertical bar is positioned to the left of the main title text.

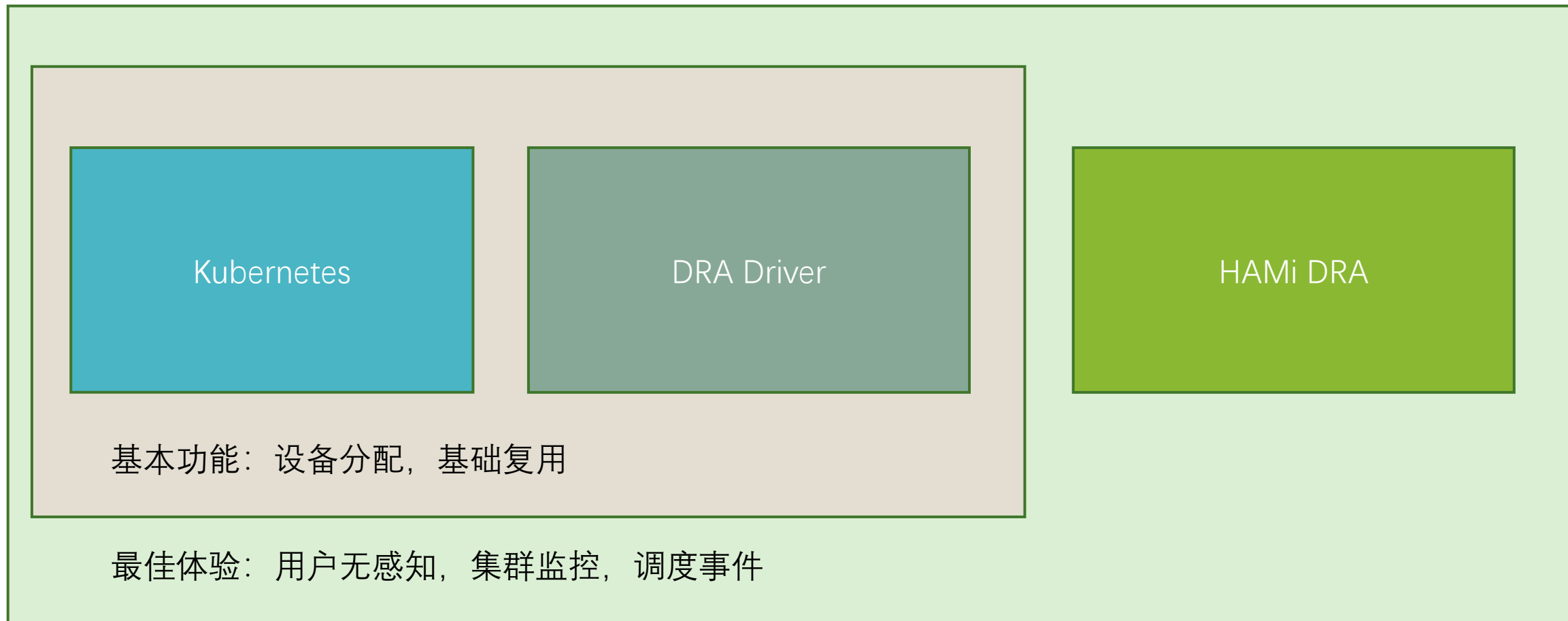
DRA 生态联盟

DRA:Dynamic Resource Allocation

KEP	Description	Owner SIG	Lead	Released			Targets			Notes
	See also the DRA enhancements project board .			1.32	1.33	1.34	1.35	1.36	1.37	
5004	DRA: Extended Resource Requests via DRA	Scheduling	yliaog			Alpha	Alpha	Beta		Avoid user needing to know device plugin vs DRA
4817	DRA: Resource Claim Status	Node	LionelJouin	Alpha	Beta	Beta	Beta	Stable		Enables multi-network services
5018	DRA: Namespace Controlled Admin Access	Auth	ritazh		Alpha	Beta	Beta	Stable		No changes in 1.35, GA in 1.36
5055	DRA: Device Taints and Tolerations	Scheduling	pohly		Alpha	Alpha	Alpha	Beta		Health information, evict pods
4815	DRA: Partitionable Devices	Scheduling	mortent		Alpha	Alpha	Alpha	Beta		Underpins dynamic MIG/TPU, TPU slice allocation
4680	Resource Health Status in Pod Status	Node	jpsassine	Alpha	Alpha	Alpha	Beta?	Stable?		Inform users about everything affecting their pods
5007	DRA: Device Binding Conditions	Scheduling	ttsuuubasa			Alpha	Alpha	Beta		Dynamically attach devices to nodes
5075	DRA: Consumable Capacity	Scheduling	sunya-ch			Alpha	Alpha	Beta		Basis for platform-mediated device sharing
4816	DRA: Prioritized Alternatives in Device Requests	Scheduling	mortent		Alpha	Beta	Beta	Stable		Second beta to add scoring
5491	DRA: List Types for Attributes	Scheduling	bg-chun, everpeace					Alpha		More flexible hardware topologies
5234	DRA: ResourceSlice Mixins	Scheduling	mortent					Alpha		Makes resource slices more compact
5194	DRA: ReservedFor Workloads	Scheduling	mortent					Alpha		Enable TPU slice scale > 256 nodes
5517	DRA: Native Resource Requests	Node	pravk03					Alpha		
5677	DRA: Resource Availability Visibility									UX improvement
5676	DRA: Explicit DeviceClass Fields	Scheduling								Performance optimization, especially for WAS
	Early Ideas									
	WAS integration	Scheduling								Probably part of the WAS KEP
	Likely something for contiguous group allocations	Scheduling	aws-ruhip							
	Likely something for a provisioning flow	Scheduling	johnbelamaric							
	Maybe connect consumable and partitionable	Scheduling	johnbelamaric							
	Maybe something for topological allocations	Scheduling	johnbelamaric							



HAMi DRA解决方案



HAMi的优势

01

CNCF是一个开放的社区，
Kubernetes是一个很包容的项目。

02

CNCF的市场部门
对于中国市场非常重视。

03

Kubernetes社区
需要国内异构设备厂商的支持。

04

HAMi社区以及密瓜智能的中立性和专
业性受到了厂商们的广泛认可。

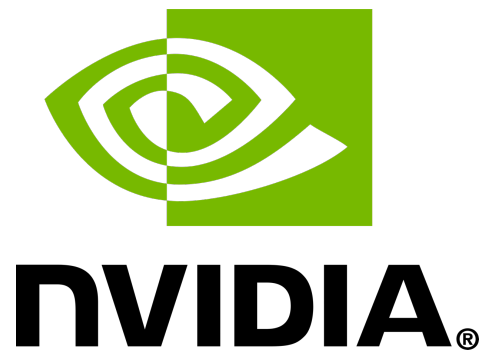
05

HAMi社区会在国内帮助推广
Kubernetes的新版本/新功能。

06

HAMi已被大量的企业采用
在云原生和AI领域有一定影响力和认可度

第一批HAMi支持的DRA生态



THANK YOU